

Sun's 2-4-6-8 Binomial Conjecture: An Audited Circle Method Framework and Obstruction Analysis

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¹ Zenodo Preprints V2.0 & V3.1 (July 23, 2026) | ² Repository: <https://github.com/Raj123-0/bounty-math-solver>

Credit Line: This report compiles and extends research originally presented by Scott Sun in his Zenodo preprints (2026) [1,2].

Abstract

In 2019, Zhi-Wei Sun conjectured that every positive integer $n \geq 1$ can be expressed as $n = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$ for non-negative integers $w, x, y, z \in \mathbb{N}$. Following the analytical framework of Scott Sun (2026), this report establishes the Hardy–Littlewood circle method framework for this representation problem. We prove the exact 2-adic local density $\chi_2(n) = 1.000000$ identically and establish an unconditional positive lower bound $\mathfrak{S}(n) \geq c_0 > 0$ for the singular series. The continuous singular integral is derived rigorously step by step via the Dirichlet–Liouville integral:

$$J(n) = c_J n^{1/24}, \quad c_J = \frac{\Gamma(\frac{1}{2})\Gamma(\frac{1}{4})\Gamma(\frac{1}{6})\Gamma(\frac{1}{8})}{\Gamma(\frac{25}{24})} \prod_{k \in \{2,4,6,8\}} \frac{(k!)^{\frac{1}{k}}}{k} \approx 25.323984,$$

which is independently validated against direct 3D adaptive numerical quadrature (relative error $< 2 \times 10^{-7}$). A conditional reduction theorem is proven: if the minor arc estimate $\int_{\mathfrak{m}} |f_2 f_4 f_6 f_8| d\alpha \ll N^{1/24-\eta}$ holds for some $\eta > 0$, then Sun's conjecture follows for all sufficiently large n .

Conversely, a comprehensive obstruction analysis demonstrates why seven standard analytic techniques fail by exponent deficits ranging from 0.3958 to 0.7292. The critical dimension $\sum_{k \in \{2,4,6,8\}} \frac{1}{k} = \frac{25}{24}$ barely exceeds 1, causing all known methods to fall short. Windowed statistical averages against OEIS A306477 counts up to $N = 75,000$ show that the empirical ratio $\frac{\bar{r}}{\mathfrak{S}_J}$ drifts from 0.72 to 1.33 (a $\approx 1.85 \times$ spread), indicating missing lower-order polynomial terms. Complete Python verification code and terminal outputs are provided in Appendices A–C. The main conjecture remains open.

1 Introduction

In 2019, Zhi-Wei Sun formulated a collection of conjectures on mixed sums of binomial coefficients and polynomial sequences. A prominent problem in this collection, cataloged as OEIS A306477, states:

Conjecture 1.1 (Sun's 2-4-6-8 Binomial Sum Conjecture). Every positive integer $n \geq 1$ can be represented in the form:

$$n = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$$

for non-negative integers $w, x, y, z \in \mathbb{N} = \{0, 1, 2, \dots\}$.

Conjecture 1.1 has been verified computationally without counterexamples for all $n \leq 2 \times 10^{12}$ by Yaakov Baruch (OEIS A306477, March 2019). The representation function is defined as:

$$r(n) = \left| \left\{ (w, x, y, z) \in \mathbb{N}^4 : n = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8} \right\} \right|.$$

1.1 Methodological Status and Proven vs. Heuristic Classification

The main conjecture remains OPEN. Throughout this report, we maintain a strict separation between:

- **Proven Theorems:** exact p -adic density for $p = 2$, singular series positivity $\mathfrak{S}(n) \geq c_0 > 0$, singular integral evaluation $J(n)$, conditional reduction to minor arcs, and obstruction deficit bounds.
- **Numerically Determined Results:** local densities for $p = 3, 5, 7$ (with code in Appendix B) and windowed statistical averages (Appendix C).
- **Heuristic Estimates:** standard circle method major arc approximations originating in Scott Sun (2026).

2 Preliminaries and Uniform Notation

For non-negative integers $w, x, y, z \geq 0$, define the integer-valued binomial polynomials:

$$P_2(w) = \binom{w+2}{2} = \frac{(w+2)(w+1)}{2}, \quad P_4(x) = \binom{x+3}{4} = \frac{(x+3)(x+2)(x+1)x}{24},$$

$$P_6(y) = \binom{y+5}{6} = \frac{(y+5)(y+4)\dots y}{720}, \quad P_8(z) = \binom{z+7}{8} = \frac{(z+7)(z+6)\dots z}{40320}.$$

Notice that $P_2(0) = 1$, while $P_4(0) = P_6(0) = P_8(0) = 0$. Consequently, $(w, x, y, z) = (0, 0, 0, 0)$ represents $n = 1$.

Let $N \geq 1$ be a large scaling parameter. For each $k \in \{2, 4, 6, 8\}$, define the scaling limits $X_k = (k!N)^{\frac{1}{k}}$ and the generating Weyl exponential sums:

$$f_{k(\alpha)} = \sum_{0 \leq u \leq X_k} e(\alpha P_k(u)) \quad \left(\alpha \in \mathbb{T} = \frac{\mathbb{R}}{\mathbb{Z}} \right),$$

where $e(t) = \exp(2\pi it)$. By character orthogonality, the representation number $r(n)$ for $n \leq N$ is given by the Fourier integral:

$$r(n) = \int_0^1 f_2(\alpha) f_4(\alpha) f_6(\alpha) f_8(\alpha) e(-n\alpha) d\alpha.$$

2.1 Major and Minor Arc Dissection

Let $\theta = \frac{1}{24}$. For integers $1 \leq a \leq q \leq N^\theta$ with $\gcd(a, q) = 1$, define the major arcs:

$$\mathfrak{M}(q, a) = \left\{ \alpha \in [0, 1) : \left| \alpha - \frac{a}{q} \right| \leq \frac{1}{qN^{1-\theta}} \right\}, \quad \mathfrak{M} = \bigcup_{q \leq N^\theta} \bigcup_{(a,q)=1} \mathfrak{M}(q, a).$$

The minor arcs are defined as the complement $\mathfrak{m} = [0, 1) \setminus \mathfrak{M}$.

3 The Major Arcs: Rigorous Local Analysis

3.1 The Singular Integral: Complete Step-by-Step Derivation

On the major arcs, the leading continuous oscillatory integral is $v_{k(\beta)} = \int_0^{(k!N)^{\frac{1}{k}}} e(\beta \frac{t^k}{k!}) dt$.

Lemma 3.1 (Inverse Fourier Transform of v_k). For each $k \in \{2, 4, 6, 8\}$ and $\tau \in [0, N]$, the inverse Fourier transform of $v_{k(\beta)}$ is the fractional power density:

$$\varphi_{k(\tau)} := \mathcal{F}^{-1}[v_k](\tau) = \frac{(k!)^{\frac{1}{k}}}{k} \tau^{\frac{1}{k}-1}.$$

Theorem 3.2 (Proven: Continuous Singular Integral). The continuous singular integral evaluates explicitly for $n > 0$ to:

$$J(n) = \frac{\Gamma(\frac{1}{2})\Gamma(\frac{1}{4})\Gamma(\frac{1}{6})\Gamma(\frac{1}{8})}{\Gamma(\frac{25}{24})} \left(\frac{2^{\frac{1}{2}}}{2} \cdot \frac{24^{\frac{1}{4}}}{4} \cdot \frac{720^{\frac{1}{6}}}{6} \cdot \frac{40320^{\frac{1}{8}}}{8} \right) n^{1/24} = c_J n^{1/24},$$

where $c_J \approx 25.323984$.

Erratum & History of the Constant: In earlier preprints (such as Scott Sun 2026), $c_J \approx 0.02035$ was reported due to applying $\Gamma(1 + \frac{1}{k}) = \frac{1}{k}\Gamma(\frac{1}{k})$ instead of $\Gamma(\frac{1}{k})$ and inverting coefficients. The correct constant $c_J = 25.323984$ was verified against independent 3D numerical quadrature (relative error $< 2 \times 10^{-7}$; code in Appendix A).

3.1.1 Numerical Validation of $J(n)$

n	Direct Numerical $J_{\text{num}}(n)$	Analytical Formula $c_J n^{1/24}$	Relative Difference
10	27.873956	27.873962	1.98×10^{-7}
50	29.807288	29.807287	3.89×10^{-8}
100	30.680708	30.680707	1.15×10^{-8}

Table 1: Independent 3D Numerical Quadrature vs. Analytical Formula $c_J n^{\frac{1}{24}}$ (Data generated via adaptive numerical integration in Python; theoretical formula from Scott Sun (2026) with corrected normalization).

3.2 Rigorous Computation of Local Densities

For each prime p and integer $k \geq 1$, define $M(p^k, n) = |\{(w, x, y, z) \bmod p^k : P_2(w) + P_4(x) + P_6(y) + P_8(z) \equiv n \bmod p^k\}|$ and the p -adic local density $\chi_{p(n)} = \lim_{k \rightarrow \infty} p^{-3k} M(p^k, n)$.

Theorem 3.3 (Proven: Exact 2-adic Equidistribution). For all $n \geq 1$, $\chi_2(n) = 1.000000$ identically, because $2P_{2'}(w) = 2w + 3 \equiv 1 \bmod 2 \in \mathbb{Z}_2^\times$ is a 2-adic unit.

Prime p	Period Modulo	Min $\chi_{p(n)}$	Max $\chi_{p(n)}$	Characteristics
3	9	0.765432	1.111111	Minimum at $n \equiv 5 \pmod 9$; mean = 1.0
5	25	0.889600	1.120000	Minimum at $n \equiv 19 \pmod{25}$; mean = 1.0
7	49	0.903790	1.131195	Minimum at $n \equiv 19, 33, 40 \pmod{49}$

Table 2: Small Prime Local Densities (Numerically determined via discrete FFT convolution on $\frac{\mathbb{Z}}{p^2}\mathbb{Z}$; methodology adapted from Scott Sun (2026)).

Theorem 3.4 (Weil Bound for $p > 7$). For all primes $p > 7$, $|\chi_{p(n)} - 1| \leq \frac{105}{p^2}$. Consequently, $\mathfrak{S}(n) \geq c_0 \approx 0.1218 > 0$.

3.3 Conditional Reduction to the Minor Arc Estimate

Theorem 3.5 (Proven: Conditional Proof of Sun's Conjecture). Assume the **Joint Minor Arc Estimate (J)**: there exist constants $\eta > 0$ and $C > 0$ such that:

$$\int_{\mathfrak{m}} |f_2(\alpha)f_4(\alpha)f_6(\alpha)f_8(\alpha)| d\alpha \leq C \cdot N^{1/24-\eta}.$$

Then under standard major arc asymptotics, Sun's conjecture holds for all sufficiently large $n \geq n_0$.

4 The Obstruction Analysis: Survey of the Standard Analytic Toolkit

The central obstruction identified by Scott Sun [1] arises because the critical dimension

$$\sum_{k \in \{2,4,6,8\}} \frac{1}{k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} = \frac{25}{24} \approx 1.041667$$

barely exceeds 1. Below, we review seven standard analytic paradigms and compute their exact exponent deficits $\Delta = \text{Exponent} - \frac{1}{24}$:

1. **Obstruction 1 (Cauchy-Schwarz & Mean-Value):** Uncoupling f_8 on minor arcs gives $|f_8|_{L^2} = N^{1/16}$. Under heuristic mean-value $N^{5/12}$, this gives $N^{23/48}$ ($\Delta = 0.4375$). Under rigorous Hua $N^{17/24}$, this gives $N^{37/48}$ ($\Delta = 0.7292$).
2. **Obstruction 2 (Multi-Dim. Weyl Differencing):** The form is diagonal. Linearizing P_8 requires 7 differences ($2^7 = 128$), but after difference 2, $\Delta^2 P_2 \equiv 0$, yielding trivial volume X_2^{128} and collapsing to $N^{23/48}$ ($\Delta = 0.4375$).
3. **Obstruction 3 (4D Poisson Summation):** Boundary error dominated by $w \leq X_2 = N^{1/2}$ yields $\frac{X_2 X_4 X_6 X_8}{X_2} = N^{13/24}$ ($\Delta = 0.5000$).
4. **Obstruction 4 (Farey / Kloosterman Refinement):** Weil bound $|S(p, a)| \leq 105p^2$ summed over $q \leq N^{1/2}$ gives $\mathcal{O}(N^{7/16})$ ($\Delta = 0.3958$).
5. **Obstruction 5 (BDG Decoupling on 1D Slice):** Hölder exponents $(4, 8, 12, 24)$ on single-variable moments yield $N^{\frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{24}} = N^{1/2}$ ($\Delta = 0.4583$).
6. **Obstruction 6 (Deligne Finite Field Bounds):** Degeneration to non-isolated singular locus $z^8 = 0$ at infinity forces $A(p, n) = \mathcal{O}(p^{-1})$, summing to $\log N$ (diverges).
7. **Obstruction 7 (Igusa p -Adic Zeta Functions):** Log-canonical threshold $\text{lct}(f, \mathbf{0}) = 1$ locks the dominant pole at $s = -1$, proving $|A(p, n)| \asymp p^{-1}$ is sharp (diverges).

Method	Analytical Exponent	Numerical Exponent	Exponent Deficit Δ
Target Main Term ($J(n)$)	$N^{1/24}$	+0.0417	0.0000
1. Cauchy–Schwarz (Heuristic)	$N^{1/16}N^{5/12} = N^{23/48}$	+0.4792	+0.4375
1'. Cauchy–Schwarz (Rigorous Hua)	$N^{1/16}N^{17/24} = N^{37/48}$	+0.7708	+0.7292
2. Multi-Dim. Weyl Differencing	$N^{23/48}$ (collapses)	+0.4792	+0.4375
3. 4D Poisson Boundary Error	$N^{13/24}$	+0.5417	+0.5000
4. Farey / Kloosterman Refinement	$N^{7/16}$	+0.4375	+0.3958
5. BDG ℓ^2 1D Slice Decoupling	$N^{1/2}$	+0.5000	+0.4583
6. Deligne Finite Field Bounds	$p^{-1} \implies \mathcal{O}(\log N)$	Diverges	N/A
7. Igusa p -Adic Zeta Functions	$\text{lct} = 1 \implies p^{-1}$	Diverges	N/A

Table 3: Summary of Obstruction Exponents and Deficits over the Target $N^{\frac{1}{24}}$ (Analytical derivations from Scott Sun (2026), re-tabulated with exact fractional and decimal deficits).

5 Numerical Verification and Honest Statistical Analysis

Window $[N, 1.5N]$	Sample Size	$\overline{r(n)}$	σ_r	$\overline{\mathfrak{S}J}$	$\sigma_{\mathfrak{S}J}$	Window Ratio
[100, 150]	51	11.49	3.27	15.85	4.64	0.7248
[500, 750]	251	17.30	4.91	17.09	5.24	1.0123
[1000, 1500]	501	18.09	5.41	17.65	5.37	1.0248
[5000, 7500]	2501	22.36	7.09	18.85	5.69	1.1866
[10000, 15000]	2501	24.15	7.82	19.39	5.80	1.2452
[25000, 37500]	6251	26.89	8.44	20.15	6.04	1.3345
[50000, 75000]	5001	29.27	9.18	22.23	6.39	1.3164

Table 4: Windowed Statistical Comparison: Empirical $r(n)$ vs. Leading Asymptotic $\mathfrak{S}(n).J(n)$ (Empirical counts from OEIS A306477; asymptotic model from Scott Sun (2026) with corrected c_J).

Honest Analysis of Statistical Behavior:

- 1. **The Ratio Does NOT Stabilize Tightly:** The window ratio $\frac{\overline{r}}{\overline{\mathfrak{S}J}}$ drifts upward from 0.7248 at $N = 100$ to 1.3345 at $N = 25,000$ (a $\approx 1.85 \times$ spread) before leveling off around 1.32.
- 2. **Cause of Upward Drift:** Lower-order polynomial terms in $P_{k(u)} = \binom{u+k}{k} > \frac{u^k}{k!}$ contribute positive offsets over pure monomials at moderate ranges $N \leq 75,000$.
- 3. **Pointwise Fluctuations:** Individual representation counts fluctuate significantly ($\sigma_r \approx 3 - 9$) due to local residue sensitivity in $\chi_{p(n)}$.

6 Conclusion & The Precise Roadmap

Sun’s 2-4-6-8 conjecture is conditionally established modulo the Joint Minor Arc Estimate (J). To establish the conjecture unconditionally, research must focus on either:

- 1. **Sub- p^{-1} Cohomological Cancellation:** $|A(p, n)| \leq Cp^{-1-\delta}$ for some $\delta > 0$.
- 2. **1D Mixed Curve Decoupling:** $\int_{\mathfrak{m}} |f_2(\alpha)f_4(\alpha)f_6(\alpha)f_8(\alpha)| \, d\alpha \leq CN^{1/24-\eta}$ for some $\eta > 0$.

Editor’s Note: This document compiles and extends research originally presented by Scott Sun in his Zenodo preprints (2026) [1,2]. This revision corrects the exponent in Theorem 3.5 and Section 6 from n to η , fixes algebraic errors in the obstruction table, standardizes 2-adic notation to $\mathbb{Z}_2^\times$, merges duplicate sections, and clarifies table captions. Appendices containing code and extended data are available in the original Zenodo preprints [1,2] and the companion repository.

References:

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2. S. Sun, *Representation Geometry and Structural Bridges in Binomial Representation Systems*, Zenodo Preprint (Version V3.1, July 23, 2026).
3. OEIS Foundation Inc., *Sequence A306477: Number of representations of n as $\binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$* , The On-Line Encyclopedia of Integer Sequences (Conjectured by Z.-W. Sun, verified up to 2×10^{12} by Y. Baruch, 2019), <https://oeis.org/A306477>.
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5. J. Bourgain, C. Demeter, and L. Guth, *Proof of the main conjecture in Vinogradov's Mean Value Theorem*, Ann. of Math. **184** (2016), 633–682.
6. P. Deligne, *La conjecture de Weil. I*, Publ. Math. IHÉS **43** (1974), 273–307.
7. J. Denef, *The rationality of the Poincaré series*, Invent. Math. **77** (1984), 1–23.
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Appendix A:

Python Code: Direct Numerical Quadrature of $J(n)$

```
import math
import numpy as np
import scipy.integrate as integrate

def numerical_quadrature_J(n: float) -> float:
    # Direct 3D adaptive numerical integration with regularized coordinates
    c2 = 2**0.5 / 2.0
    c4 = 24**0.25 / 4.0
    c6 = 720**(1/6) / 6.0
    c8 = 40320**0.125 / 8.0

    # Regularized integrand eliminating endpoint singularities: t1=x^2, t2=y^4, t3=z^6
    def reg_integrand(z, y, x):
        rem = n - x**2 - y**4 - z**6
        if rem <= 1e-12: return 0.0
        return 2.0 * 4.0 * 6.0 * (rem**(-7/8))

    x_max = n**0.5
    def y_max(x): return max(0.0, n - x**2)**0.25
    def z_max(x, y): return max(0.0, n - x**2 - y**4)**(1/6)

    val, _ = integrate.tplquad(
        reg_integrand, 0, x_max,
        lambda x: 0, y_max,
        lambda x, y: 0, z_max,
        epsabs=1e-5, epsrel=1e-5
    )
    return c2 * c4 * c6 * c8 * val

if __name__ == "__main__":
    cJ = (math.gamma(0.5)*math.gamma(0.25)*math.gamma(1/6)*math.gamma(0.125)/math.gamma(25/24)) *
    (2**0.5/2.0) * (24**0.25/4.0) * (720**(1/6)/6.0) * (40320**0.125/8.0)
    for n in [10, 50, 100]:
        num = numerical_quadrature_J(n)
        theo = cJ * (n ** (1/24))
        print(f"n={n:3d}: numeric={num:.6f}, theoretical={theo:.6f}, rel diff={abs(num-theo)/theo:.2e}")
```

Terminal Output:

```
n= 10: numeric=27.873956, theoretical=27.873962, rel diff=1.98e-07
n= 50: numeric=29.807288, theoretical=29.807287, rel diff=3.89e-08
n=100: numeric=30.680708, theoretical=30.680707, rel diff=1.15e-08
```

Appendix B:

Python Code: Small Prime Local Density Engine

```
import numpy as np

def compute_local_density_table(p: int, k: int) -> np.ndarray:
    mod = p ** k
    def get_poly_hist(poly_func, max_deg):
        period = mod * max_deg
        vals = [poly_func(w) % mod for w in range(period)]
        hist = np.zeros(mod, dtype=np.float64)
        for v in vals:
            hist[v] += 1.0
        return hist / len(vals)

    h2 = get_poly_hist(lambda w: (w+2)*(w+1)//2, 2)
    h4 = get_poly_hist(lambda x: (x+3)*(x+2)*(x+1)*x//24, 4)
    h6 = get_poly_hist(lambda y: (y+5)*(y+4)*(y+3)*(y+2)*(y+1)*y//720, 6)
    h8 = get_poly_hist(lambda z: (z+7)*(z+6)*(z+5)*(z+4)*(z+3)*(z+2)*(z+1)*z//40320, 8)

    # 4-fold discrete convolution via FFT
    f2, f4, f6, f8 = np.fft.fft(h2), np.fft.fft(h4), np.fft.fft(h6), np.fft.fft(h8)
    conv = np.real(np.fft.ifft(f2 * f4 * f6 * f8)) * mod
    return conv

if __name__ == "__main__":
    for p, k in [(3, 2), (5, 2), (7, 2)]:
```

```
t = compute_local_density_table(p, k)
print(f"p={p}, mod {p**k}: min={np.min(t):.6f}, max={np.max(t):.6f}, mean={np.mean(t):.6f}")
```

Terminal Output:

```
p=3, mod 9: min=0.765432, max=1.111111, mean=1.000000
p=5, mod 25: min=0.889600, max=1.120000, mean=1.000000
p=7, mod 49: min=0.903790, max=1.131195, mean=1.000000
```

Appendix C:

Python Code: Windowed Statistical Verification

```
import numpy as np
from src.a306477_checker import count_representations
from src.local_density_calculator import compute_local_density_table, compute_singular_series

primes_tables = {p: compute_local_density_table(p, 2 if p <= 7 else 1)
                  for p in [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31]}
tail = np.prod([1.0 - 105.0/(p*p) for p in range(37, 5000)
                if all(p % d != 0 for d in range(2, int(p**0.5)+1))])
cJ = 25.323984

def compute_window(n_start: int, n_end: int, step: int = 1):
    ns = list(range(n_start, n_end + 1, step))
    actual = np.array([count_representations(n) for n in ns], dtype=np.float64)
    pred = np.array([compute_singular_series(n, primes_tables, tail) * cJ * (n**(1/24)) for n in ns])
    return len(ns), np.mean(actual), np.std(actual), np.mean(pred), np.std(pred), np.mean(actual)/
    np.mean(pred)

if __name__ == "__main__":
    for N, step in [(100, 1), (500, 1), (1000, 1), (5000, 1), (10000, 2), (25000, 2), (50000, 5)]:
        cnt, m_act, s_act, m_pred, s_pred, ratio = compute_window(N, int(1.5*N), step)
        print(f"[{N:5d}, {int(1.5*N):5d}] | size={cnt:4d} | mean_act={m_act:5.2f} (std={s_act:4.2f}) | "
              f"mean_pred={m_pred:5.2f} (std={s_pred:4.2f}) | ratio={ratio:.4f}")
```

Terminal Output:

```
[ 100,   150] | size=  51 | mean_act=11.49 (std=3.27) | mean_pred=15.85 (std=4.64) | ratio=0.7248
[  500,   750] | size= 251 | mean_act=17.30 (std=4.91) | mean_pred=17.09 (std=5.24) | ratio=1.0123
[ 1000,  1500] | size= 501 | mean_act=18.09 (std=5.41) | mean_pred=17.65 (std=5.37) | ratio=1.0248
[ 5000,  7500] | size=2501 | mean_act=22.36 (std=7.09) | mean_pred=18.85 (std=5.69) | ratio=1.1866
[10000, 15000] | size=2501 | mean_act=24.15 (std=7.82) | mean_pred=19.39 (std=5.80) | ratio=1.2452
[25000, 37500] | size=6251 | mean_act=26.89 (std=8.44) | mean_pred=20.15 (std=6.04) | ratio=1.3345
[50000, 75000] | size=5001 | mean_act=29.27 (std=9.18) | mean_pred=22.23 (std=6.39) | ratio=1.3164
```